ModelIt!

Teacher's Guide

Can We Stop a Flood?

Weather Hazards and Staying Safe

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-ESS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-ESS3-2

Ask questions to obtain information about the purpose of weather forecasting to prepare for, and respond to, severe weather.

Three-Dimensional Learning

Science Practice	**Asking Questions and Defining Problems** — Students ask questions about how to protect homes from flooding and define the problem of stopping water with a barrier.
Disciplinary Core Idea	**ESS3.B Natural Hazards** — Some kinds of severe weather are more likely than others in a given region. Weather scientists forecast severe weather so that communities can prepare for and respond to these events.
Crosscutting Concept	**Cause and Effect** — Students observe that more rain (cause) leads to higher flood levels (effect), and that barriers can reduce the effect.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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> **For Your Confidence**
>
> This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Flooding is one of the most common natural hazards. Teaching young children about weather preparedness empowers them to understand that while we cannot control the weather, we can design solutions to reduce its impact on our communities.

What Causes Floods?

Floods occur when water overflows onto land that is normally dry. The most common cause is heavy rainfall that exceeds the ground's ability to absorb water. Rivers and streams can overflow their banks, and storm drains can back up in cities. Flash floods happen quickly after intense rain. Coastal areas can flood from storm surges during hurricanes. Understanding that more rain equals more flood risk is the foundational concept for kindergarteners.

How People Protect Against Floods

Humans have developed many ways to manage flood risk. Levees and dams hold back river water. Sandbags create temporary barriers during emergencies. Proper drainage systems channel water away from buildings. Retention ponds store excess water during storms. At the kindergarten level, the key concept is that BARRIERS (walls, bags, ditches) can block or redirect water to keep homes safe. This introduces engineering design thinking — identifying a problem and building a solution.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Amount of Rain as the external component (weather controls how much rain falls) and Flood Level as the internal component (it rises or falls based on how much rain comes).

E **Establish Relationships**
Students discover that more rain leads to higher Flood Level — a positive relationship.

V **Visualize & Model**
Students build a simple two-part model connecting Amount of Rain to Flood Level with a positive arrow.

E **Evaluate Outcomes**
Students test Light Rain vs. Heavy Rain scenarios on the sand table and observe flooding differences.

R **Revise & Extend**
Students explore what happens when they add a barrier — can they reduce the Flood Level even when rain is heavy?

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever seen a really big puddle after it rains? What if the puddle got SO big it covered your whole yard? That is a FLOOD!"*

Watch For

Show a picture of a small puddle next to a picture of a real flood. Ask: What is different? How did the water get so high?

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn what causes floods and how people STOP them! We will build our own flood barriers like real engineers!"*

Watch For

Introduce 'flood' and 'barrier.' Students say each word. Show a picture of sandbags and ask: What are these for?

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: Can we STOP a flood? If it rains and rains and rains, what can people build to keep the water away?"*

Watch For

Students brainstorm with a partner: What could you build to stop water? Share ideas as a class and list them on the board.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Amount of Rain is how much rain falls — the WEATHER decides that. Flood Level is how high the water gets — that depends on the rain!"*

Watch For

Use sorting mats. Students place Amount of Rain on OUTSIDE and Flood Level on INSIDE. Sprinkle water (light rain) then pour water (heavy rain) into a shallow pan.

Slide 5: How Are They Connected?

5 min

Say: *"When MORE rain falls, the Flood Level goes UP! That is a POSITIVE connection. More rain means more flooding."*

Watch For

Demonstrate on the sand table: pour a little water — small puddle. Pour a lot of water — flood! Students give thumbs up for positive.

Slide 6: What Did We Find Out?

5 min

Say: *"More rain means more flooding. But smart engineers build BARRIERS to block the water! Sandbags, walls, and levees protect homes."*

Watch For

Chant: 'When Rain goes up, Flooding goes up!' Complete the sentence frame together. Discuss: What can we BUILD to help?

Slide 7: STEM Challenge

10 min

Say: *"A big storm is coming to Toy Town! Can your team build a barrier to stop the flood? Test it by pouring water!"*

Watch For

Distribute sand table supplies: clay, craft sticks, foil, plastic wrap, toy houses. Groups build barriers. Test by slowly pouring water and observing. Discuss what worked and what did not.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Amount of Rain
- **INTERNAL:** Flood Level

Relationships

- **Amount of Rain to Flood Level:** POSITIVE (+) — When more rain falls, the flood level gets higher because the ground cannot absorb all the water and it pools on the surface.

Key Discoveries

- More rain makes higher flood levels — lots of rain can cover roads and homes
- Barriers like walls and sandbags can slow down or stop water from flooding
- People can prepare for floods by building barriers and moving to higher ground

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Floods only happen near the ocean"	Floods can happen anywhere it rains heavily. Rivers, streams, and even streets in cities can flood when too much rain falls too quickly. Floods happen in mountains, plains, deserts, and cities — not just near the ocean.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"You can stop all floods with a wall"	While barriers help reduce flooding, very heavy rains or huge storms can overwhelm even strong barriers. That is why weather forecasting is so important — knowing a big storm is coming lets people prepare, evacuate, and stay safe even when barriers are not enough.